Leonardo Matteo Bolognese

leonardo.bolognese@epfl.ch | github.com/utopy77 | linkedin.com/in/leonardo-matteo-bolognese/

Education

EPFL *Sep 2026 - Jul 2028*
MSc | *Data Science* *Lausanne, CH*

EPFL *Sep 2023 - Jul 2026*
BSc | *Communication Systems* *Lausanne, CH*

Skills

Expertise: Machine Learning, Statistics, Data Engineering, NLP, Algorithms, Data Structures, Signal & Image Processing, Digital Communications, Cryptography & Security, Embedded Systems & Control (PID), Computer Networks, Data-intensive Systems
Languages: Python, C/C++, Scala, SQL, Java, Verilog
Tools: NumPy / pandas / scikit-learn, Selenium, PostgreSQL, Git, PlatformIO, LaTeX / Typst
Languages (spoken): Italian (native), English (Fluent), French (B1)

Experience

Software Developer (Data Engineering) *Mar 2024 - Sep 2025*
Agon Partners (formerly Konzeptplus) *Remote, Part-time*

- Developed Python data pipelines (Selenium, BeautifulSoup, pandas) to collect, parse and clean public data from Swiss institutional websites and documents
- Applied NLP and statistical methods to structure text data as input to a predictive model
- Worked on backend and database integration in a small remote team

Software Developer *Jun 2023 - Sep 2023*
Involve Space *Erba (CO), IT*

- Analysed telemetry and flight data from aerostatic balloon operations to support mission evaluation

Projects

Ratatoskr, Micromouse robot (EPFL CS-358)

- Built from scratch, in a team of 5, an autonomous micromouse (custom 3D-printed chassis, ESP32, time-of-flight and gyro sensing) that explores a maze and then runs the fastest path
- Implemented the flood-fill maze-solving and path optimisation, and the PID control loops, tuning them for smooth and precise motion

Profiling Chicago's neighbourhoods from bike-sharing data (EPFL ADA)

- Modelled Divvy bike-sharing trips (2013-2019) as Markov chains over Chicago's 77 community areas for four time windows (rush hours, weekend, nightlife), computing stationary distributions and transition entropy
- Clustered areas on hourly net-flow and Markov features (PCA + clustering) to recover urban zones, e.g. employment centres and residential areas, without labels

Heart-disease risk prediction (EPFL CS-433 Machine Learning)

- Predicted coronary heart disease from the CDC BRFSS 2015 survey, implementing in pure NumPy gradient descent, SGD, least squares, ridge and (regularised) logistic regression
- Built domain-driven preprocessing (special-code recoding, imputation, one-hot encoding, standardisation) and a stratified k-fold CV pipeline with F1/AUC metrics, decision-threshold tuning and an ablation study

Stargazer (PoliSpace, Politecnico di Milano association)

- In a team of 5, designed a balloon-borne probe to photograph the Milky Way; implemented PID stabilisation and image centring
- Launched successfully in July 2026, now working on a second iteration

Awards

Italian CanSat 2023 winner (link), represented Italy at the European final in Granada

Italian U16 chess champion and U14 vice-champion

ESA AstroPi project finalist (document here)